

Project Design phase 2

Solution Requirements(functional & non functional)

Date	12 July 2026
Team id	
Project name	Smart Agriculture
Maximum marks	4 marks

Functional Requirements – Smart Agriculture System

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Registration	Registration through mobile number and password Registration through Gmail Farmer profile creation (name, farm location, crop details)
FR-2	User Authentication	Login using mobile number/email OTP verification Password reset via email/OTP
FR-3	Farm & Crop Management	Add farm details Add crop type and sowing date Update crop information
FR-4	Weather Monitoring	Display real-time weather updates Rainfall and temperature forecast Weather alerts and notifications
FR-5	Soil Monitoring	Display soil moisture data Soil nutrient analysis Soil health recommendations
FR-6	AI Crop Recommendation	Recommend suitable crops based on soil and weather Suggest fertilizers and irrigation schedule
FR-7	Pest & Disease Detection	Upload crop image AI-based pest and disease identification Provide treatment recommendations
FR-8	Irrigation Management	Smart irrigation alerts Water usage recommendations
FR-9	Market Price Information	Display current crop market prices Recommend the best time to sell crops
FR-10	Dashboard & Reports	View farm status, crop health, weather, and alerts Generate productivity reports

Non-Functional Requirements – Smart Agriculture System

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	The system should have a simple, user-friendly interface that is easy for farmers to use.
NFR-2	Security	User data and farm information should be protected using secure authentication and encryption.
NFR-3	Reliability	The system should provide accurate weather forecasts, crop recommendations, and sensor data consistently.
NFR-4	Performance	The application should load within 2–3 seconds and process AI recommendations quickly.
NFR-5	Availability	The system should be available 24×7 with minimal downtime.
NFR-6	Scalability	The system should support thousands of farmers and IoT devices without performance degradation.
NFR-7	Maintainability	The system should be easy to update with new AI models and agricultural features.
NFR-8	Compatibility	The application should work on Android, iOS, and modern web browsers.
NFR-9	Data Accuracy	Weather, soil, and crop recommendations should be accurate and updated in real time.
NFR-10	Backup & Recovery	User and farm data should be backed up regularly to prevent data loss.